Data Analytics for Scientific Research & Innovation

A Practical Guide Using SPSS (With R & Python Insights)

Dr. Isaac O. Ajao

School of Pure and Applied Sciences (SPAS), Federal Polytechnic, Ado-Ekiti

August 10–13, 2026

Workshop Schedule

Time	Session Focus	Key Deliverables
00:00 - 00:40	Part I: Conceptual Foundations	8 Core pre-analysis pillars
00:40 - 01:25	Part II: Hands-on SPSS Workflow	Import, Descriptives, ANOVA, Regression
01:25 - 01:45	Part III: Modern Tools (R & Python)	Open-source ecosystem, code comparison, Q&A

Section 1

Part I: Foundations Before Analysis

1. Why Researchers Need Data Analytics

Data analytics bridges the gap between raw experimental trials and publishable scientific discovery.

Core Scientific Benefits:

- Evidence-based decision making
- Distinguishing true signal from noise
- Validating experimental reproducibility
- Meeting peer-reviewed journal standards

SPAS Context: Whether quantifying cell growth (Microbiology), compound yield (Chemistry), or sensory scores (Food Science), data analytics turns numbers into evidence.

2. Understanding Scientific Data

Scientific data is rarely clean out of the instrument or notebook.

Characteristics of Experimental Data

- **Structure:** Wide format (variables as columns) vs. Long format.
- **Quality:** Presence of missing values, human entry errors, or machine drift.
- **Sample Size (n):** Lab trials often deal with small samples ($n < 30$), requiring careful distributional checks.

Golden Rule: Clean and structure your data before attempting any statistical software analysis.

3. Variables and Measurement Scales

Your measurement scale strictly dictates which statistical test is mathematically valid.

The 4 Measurement Scales

- **Nominal (Categorical):** Unordered groups (e.g., Strain type, Department, Packaging material).
- **Ordinal (Categorical):** Ordered ranks (e.g., Likert scale, Quality grade 1–5).
- **Interval (Continuous):** Numeric without a true zero (e.g., Temperature in $^{\circ}$C).
- **Ratio (Continuous):** Numeric with a absolute zero (e.g., Weight, Yield, Concentration, Absorbance).

In SPSS, Interval and Ratio are combined into **Scale**.

4. Research Questions and Hypotheses

Every analysis answers a specific question tested through competing hypotheses:

1. Null Hypothesis (H_0)

- Statement of **no effect**, **no difference**, or **no correlation**.
- *Example:* H_0: Enzyme activity does not differ across storage temperatures.

2. Alternative Hypothesis (H_1 or H_a)

- Statement of **an effect**, **a difference**, or **a relationship**.
- *Example:* H_1: Enzyme activity differs significantly across storage temperatures.

5. Descriptive Statistics

Summarizing data characteristics before formal statistical testing.

Data Type	Central Tendency	Dispersion / Variability
Normal (Continuous)	Mean ($\bar{x}$)	Standard Deviation (s), Varianc
Skewed / Ordinal	Median	Interquartile Range (IQR)
Nominal	Mode	Frequency / Percentages (%)

Note: Reporting the Mean with skewed data can lead to misleading conclusions. Always check symmetry first.

6. Visualizing Scientific Data

Selecting the right chart for your variable type:

Categorical Data:

- **Bar Chart:** Comparing counts or percentages across groups.
- **Pie Chart:** Proportions of a whole (best for ≤ 5 groups).

Continuous Data:

- **Histogram / Density Plot:** Checking distribution shape (normality).
- **Boxplot:** Detecting outliers and checking skewness.
- **Scatter Plot:** Inspecting relationships between two variables (X vs Y).

7. Inferential Statistics & p-values

Inferential statistics allows us to draw conclusions about a **population** based on our **sample**.

The p-value (Sig. in SPSS)

- Measures the probability of observing our sample results if H_0 were true.
- **Significance Level ($\alpha = 0.05$):**
  - $p < 0.05$: Reject H_0 $\rightarrow$ Statistically significant result.
  - $p \geq 0.05$: Fail to reject H_0 $\rightarrow$ Insufficient evidence.

Effect Size (R^2, η^2)

- p-value tells you *if* there is an effect; effect size tells you *how large* or *practically meaningful* the effect is.

8. Selecting Statistical Tests (Decision Matrix)

Goal	**Input (X)**	**Output (Y)**	**Parametric**	**Non-Parametric**
Compare 2 groups	Nominal (2)	Scale (Normal)	Indep. t-test	Mann-Whitney U
Compare paired	Nominal (Paired)	Scale (Normal)	Paired t-test	Wilcoxon signed
Compare 3+ groups	Nominal (3+)	Scale (Normal)	One-Way ANOVA	Kruskal-Wallis
Relationship	Scale	Scale	Pearson r	Spearman ρ
Prediction	Scale/Nominal	Scale	Linear Regression	Logistic (Categorical Y)

Section 2

Part II: Hands-on Workflow in SPSS

Step 1: Setting Up Variable View & Data Import

In SPSS, start by defining properties correctly in **Variable View**:

1. **Name:** Short, clear variable names (`Dept`, `Temp`, `Activity`).
2. **Type:** *Numeric* for measurement readings.
3. **Values:** Label codes for categorical groups (e.g., `1 = Microbiology`, `2 = Chemistry`, `3 = Food Science`).
4. **Measure:** Explicitly set as **Scale**, **Ordinal**, or **Nominal**.

Importing Excel Files

Go to 'File > Open > Data > Files of type: Excel (*.xls, *.xlsx)'. Check "Read variable names from first row".

Step 2: Descriptives & Normality Checks

Menu Path

`Analyze > Descriptive Statistics > Explore...`

- Move outcome metric to **Dependent List**.
- Move grouping variable to **Factor List**.
- Under **Plots...**, select **Normality plots with tests**.

Interpreting Shapiro-Wilk Test

- Sig. > 0.05: Data is Normal $\rightarrow$ Proceed with Parametric Tests (ANOVA / t-test).
- Sig. < 0.05: Data is Non-Normal $\rightarrow$ Use Non-Parametric Tests (Kruskal-Wallis).

Step 3: Running One-Way ANOVA & Post-Hoc Tests

Menu Path

`Analyze > Compare Means > One-Way ANOVA...`

1. **Dependent List:** Continuous variable (e.g., `Enzyme_Activity`).
2. **Factor:** Grouping variable (e.g., `Department`).
3. **Post Hoc...:** Check **Tukey** (pinpoints *which* specific groups differ).
4. **Options...:** Check **Descriptives** & **Homogeneity of variance test**.

Reading Outputs

Check the **ANOVA** table **Sig.** column. If **Sig.** < 0.05, open the **Multiple Comparisons** table to read pairwise differences.

Step 4: Regression & Trend Prediction

Menu Path

`Analyze > Regression > Linear...`

1. **Dependent:** Variable to predict (Y).
2. **Independent(s):** Predictor variable (X).

Output Tables to Include in Publications:

- **Model Summary:** Check R **Square** (e.g., $R^2 = 0.84 \implies$ 84% variance explained).
- **ANOVA:** Check overall model **Sig.**
- **Coefficients:** Read **Unstandardized B** values to construct your predictive equation:

$$Y = \beta_0 + \beta_1(X)$$

Section 3

Part III: Beyond SPSS — R & Python

Why Explore Open-Source Tools Later?

SPSS provides immediate point-and-click results, but learning R or Python provides long-term advantages:

Key Advantages:

- **Cost**: 100% free, no license expiration.
- **Reproducibility**: Code scripts document every step.
- **Custom Graphics**: Publication-quality figures via ‘ggplot2‘.

Specializations:

- **R:** Purpose-built for advanced statistics and automated publishing (Quarto).
- **Python:** Leading language for Machine Learning and AI models.

Conceptual Code Comparison: One-Way ANOVA

1. SPSS

`Analyze > Compare Means > One-Way ANOVA...` (Dialog box execution)

2. R Language

```r
fit <- aov(Enzyme_Activity ~ Department, data = df)
summary(fit)
```

3. Python

```python
import statsmodels.api as sm
from statsmodels.formula.api import ols

model = ols('Enzyme_Activity ~ C(Department)', data=df).fit
```

Summary & Best Practices

1. **Prioritize Conceptual Clarity:** Understand measurement scales and hypotheses before clicking "Analyze".
2. **Check Assumptions:** Always verify normality before trusting p-values.
3. **Report Completely:** Include test statistics (t, F, χ^2), p-values, and effect sizes (R^2).
4. **Scale Up:** Master SPSS for immediate analysis, then transition to R or Python for advanced workflows.

Thank You! Questions & Hands-on SPSS Practice

Dr. Isaac O. Ajao | Department of Statistics, Federal Polytechnic, Ado-Ekiti